MARK SCHEME · TEACHER COPY

10BR · Chapter 5: The Internet and its Uses

Cambridge IGCSE Computer Science 0478

Paper 1 Computer Systems

Total marks: 75

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

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Question 1 · May/June 2023 Paper 13, original Q6 (a),(b),(c) [9 marks]

Question	Answer	Marks
1(a)	Any **three** from: e.g. • Checking the spelling and tone of the email/website • Checking the URL attached to a link • Scanning a download with anti-malware • Only downloading data / software from trusted sources • Never providing personal details online • Install a firewall to check if the website is valid	3
1(b)	**Two** marks for description, **one** mark for example: • Manipulating / deceiving / tricking people … • … to obtain data // to force them to make an error • Any suitable example of social engineering	3
1(c)	Any **three** from: • Providing users with different **permission** for the data • Limiting access to reading data limiting the data that can be viewed • Limiting access to editing data // limiting the data that can be deleted / changed • Normally linked to a username •	3

Question 2 · Oct/Nov 2023 Paper 12, original Q9 [6 marks]

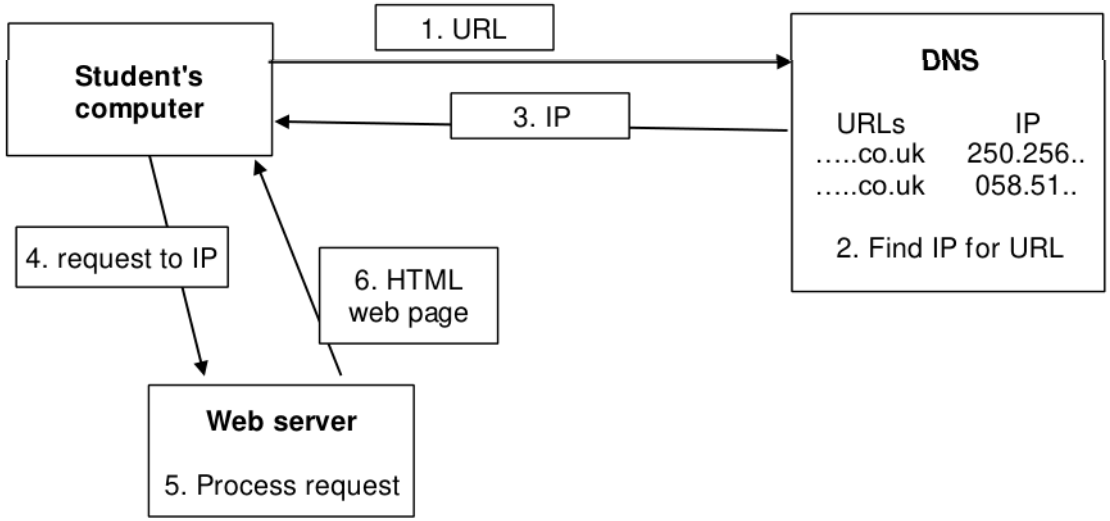
Question	Answer	Marks

2		6

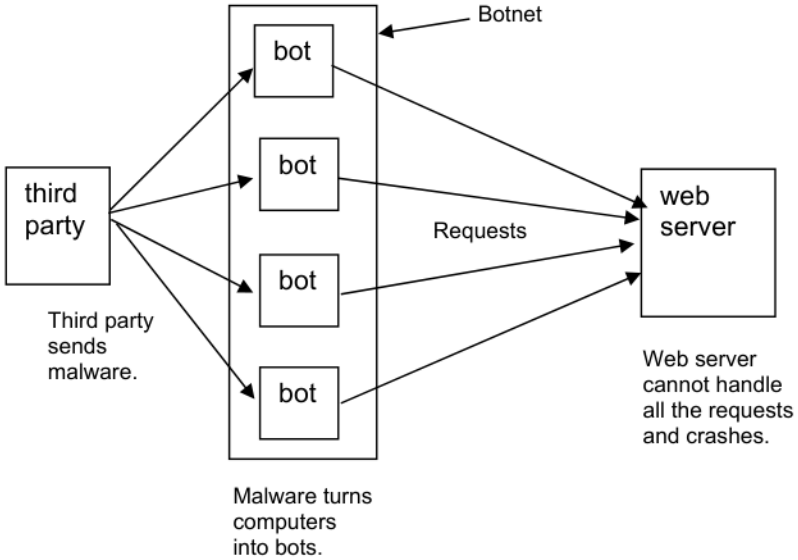
Term	Description
world wide web	the collective name for all the web pages available
cookie	a small text file, stored by the web browser, that can store a user's personal data
uniform resource locator (URL)	the **text-based address** for a website // It is made up of the protocol, domain name and filename/folder name
web server	Stores **web pages** // receives requests from clients and returns requested web page
hypertext markup language // HTML	the language used to create a website. Example tags are <head> and <body>
hypertext transfer protocol // HTTP	a protocol that is used to request and send web pages

Question 3 · Feb/March 2024 Paper 12, original Q4 [5 marks]

Question	Answer	Marks
3	Any **five** marks for each part of diagram: • **URL** from computer/browser to DNS • DNS storing table/database of **URL** and **IPs** // DNS finding **IP** for **URL** • DNS sending **IP** to computer / browser • DNS sending to higher DNS if not found • Web browser/computer sending request to **IP** of **web server** • **Web server** processing request • Web page data sent from **server** to computer / web browser	**5**



Question 4 · May/June 2024 Paper 12, original Q8 (a),(b) [6 marks]

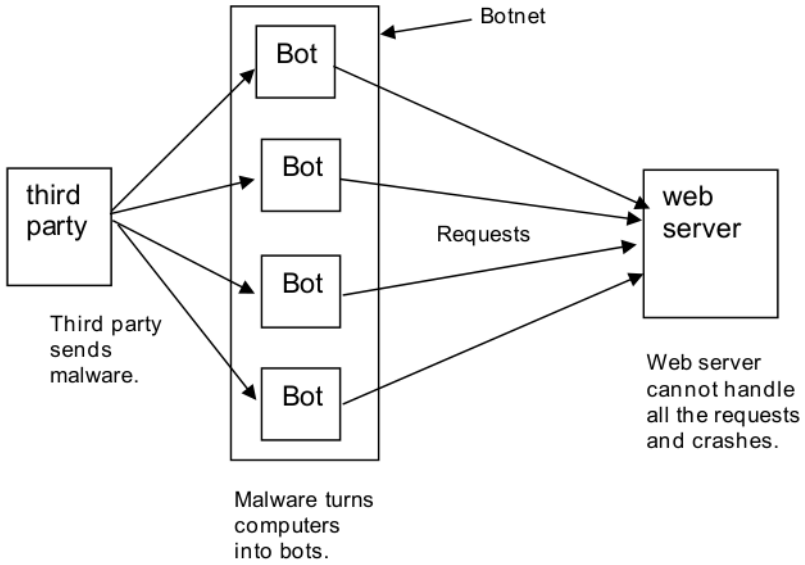
Question	Answer	Marks
4(a)	**One** mark for each part of the diagram that shows: • A perpetrator/third party sending malware // user downloads/installs malware • Each computer is turned into a bot… • … to create a botnet • Third party initiates the attack • **All** the bots send a request at once to a **web server** • … crashing the webserver Example:  	5
4(b)	Proxy server	1

Question 5 · Oct/Nov 2024 Paper 13, original Q4 [6 marks]

Question	Answer	Marks
5	**One** mark for each correct term or description in the correct place. **Term** • World wide web // WWW • IP (address) • Domain name server // DNS • Proxy server **Description** • (Web browser) **Software/application** that allows users to view/access web pages by rendering HTML • (Hacking) Gaining **unauthorised** access to a computer/system	**6**

Question 6 · May/June 2025 Paper 13, original Q5 (a),(b),(c),(d) [16 marks]

Question	Answer	Marks
6(a)	Any **two** from: • **storing** a database of URLs and matching IP addresses • **search** its database for the matching IP address. • send the URL to another DNS **if URL cannot be found.** • return the IP address to the web browser	**2**
6(b)	D	**1**

6(c)(i)	**One** mark for each part of the diagram that shows: • A perpetrator/third party sending malware // user downloads/installs malware • Each computer is turned into a bot … • … to create a botnet. • Third party initiates the attack. • **All** the bots send a request at once to the web server. • … crashing the web server. Example:  	**5**
6(c)(ii)	• Hide a public IP address • Caching	**2**
6(d)(i)	B	**1**

6(d)(ii)	Any **three** from: Examples: • Store user's login details. • Store user's payment details. • Store user's preferences. • Use for targeted advertising.	**3**
6(d)(iii)	Any **two** from: Examples: • Storing bookmarks/favourites • Recording user history • Allowing use of multiple tabs • Providing navigation tools • Providing an address bar	**2**

Question 7 · Oct/Nov 2025 Paper 12, original Q6 (a),(b),(c),(d) [14 marks]

Question	Answer	Marks
7(a)	**One** mark for each correct label • Protocol • Domain name • Web page name // file name	**3**
7(b)	Any **three** from: • It examines each request/incoming transmission • It can **slow down the rate at which/limits the number of** requests sent to the server • It can use caching … • … to respond to the request (rather than forwarding it to the server) • It can stop any requests from a certain **IP**	**3**
7(c))	**One** mark for each correct missing term or definition.	**5**

Term	Description

Total: 75 marks

	hypertext markup language (HTML)	It is the language that is often used to write/author web pages.	
	domain name server (DNS)	It is a system that stores URLs/domain names and their matching/corresponding IP addresses // It is a system that provides the matching IP address for a webpage to the web browser.	
	IP address	It is a numerical address used to locate the web server.	
	SSL/TLS/HTTPS	It is a secure protocol that can be used to transmit data sent to and from a web server.	
	web browser	It is a piece of software that is used to display web pages.	
7(d)	**One** mark for each correct missing term in the correct place. Cookies are stored and managed by the **web browser**. **Session** cookies are temporary files that are created when a user visits a web page. **Persistent** cookies are permanent files that are created when a user visits a web page.		**3**

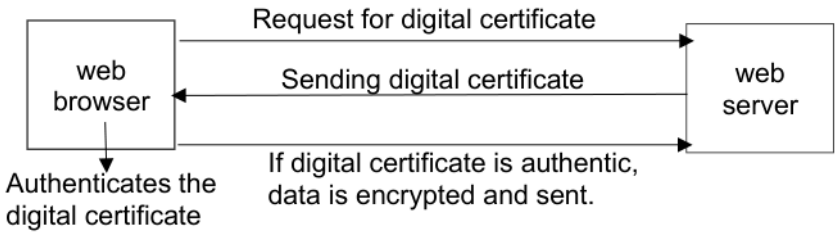
Question 8 · May/June 2026 Paper 11, original Q6 (a),(b) [10 marks]

Question	Answer	Marks	Guidance
8	Examiners must award marks according to the mark scheme and **must not** award marks with grade thresholds or grade descriptions in mind.		

Annotation	Meaning
✓	Correct point
✗	Incorrect point
?	Unclear response
FT	Follow through

REP	Repetition
I	Ignore
BOD	Benefit of doubt given
NBOD	Benefit of doubt not given
TV	Content of response too vague
NAQ	Not answered question
Annotation	**Meaning**
	Omission
	Section not relevant
	Section incorrect
Highlighter	Information copied from the text
SEEN	Page or response seen by examiner

8(a)	**One** mark for each correct term in the correct place. • domain name server (DNS) • IP address • web browser • request • hypertext markup language(HTML) The user types the URL into the address bar. The web browser sends the domain name from the URL to the **domain name server (DNS)**. This searches for the matching **IP address** and sends this back to the **web browser**. The web browser sends a **request** to the web server for the web page. The web server then sends the data for web page back to the web browser. The web browser renders the **hypertext markup language (HTML)** to display the web page.	**5**	

8(b)	**Five from, one** mark for each correct section of the diagram. • Web browser sending a request to the web server for its (digital) certificate/to identify itself • Web server sending (digital) certificate to the web browser • **Web browser authenticating/validating** the (digital) certificate … • … showing the data sent if it is valid // showing data not sent if it isn't valid • The data is **encrypted // Encryption** is used … • … using **public** and **private** keys // using **asymmetric** encryption • … to make any data sent meaningless • The **web browser** uses the **public key** to **encrypt** the data • The **web server** uses the **private key** to **decrypt** the data For example:  	**5**	MP3 - Checking the certificate/web server is safe is TV. MP3, MP4 and MP5 can be awarded in one annotation e.g. If the web browser thinks the certificate is authentic the encrypted data is sent. MP4 – BOD 'communication begins if certificate is authentic' in place of the data being sent. MP4 – Data could be sent either way, so from browser to server or server to browser MP6 – BOD two **different** keys are used, just 2 keys is TV

Question 9 · May/June 2026 Paper 13, original Q5 (c) [3 marks]

Question	Answer	Marks	Guidance
9	Examiners must use the **full range of marks** defined in the mark scheme, unless limited by the quality of the candidate responses seen.		
9(a)(i)	Any **one** from: • history • address bar	**1**	
9(a)(ii)	Any **two** from: • protocol • domain name • web page/file name // path	**2**	